

## DONGSHA DAO, Taiwan

WMO: 597920

Lat: 20.701N

Lon: 116.730E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 17.3       | 18.1 | 11.7                          | 8.6 | 18.4 | 13.3 | 9.5 | 19.0 | 19.3                  | 21.4 | 14.9 | 21.5 | 8.9                   | 50   | 0.508 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 4.5                    | 33.7            | 28.9 | 33.1 | 28.7 | 32.5 | 28.5 | 30.1                | 32.0 | 29.3 | 31.9 | 28.9 | 31.7 | 3.0                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 29.6                            | 26.6 | 31.2 | 28.7 | 25.2 | 31.1 | 28.2 | 24.4 | 30.8 | 100.3         | 31.9 | 96.6 | 32.1 | 94.5 | 31.8 | 33.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 13.3              | 11.8 | 10.5 | 15.9                       | 35.1 | 1.4                | 1.3 | 14.9                                               | 36.0 | 14.1       | 36.8 | 13.3       | 37.5 | 12.3       | 38.4 |
|                   |      |      | 12.8                       | 30.2 | 2.1                | 1.1 | 11.3                                               | 31.0 | 10.0       | 31.7 | 8.8        | 32.3 | 7.3        | 33.1 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 26.3   | 21.7  | 22.4  | 24.1  | 26.5  | 28.3  | 29.4  | 29.7  | 29.5  | 28.8  | 27.2  | 25.3  | 22.6 |
|                                                                               | DBStd       | 3.29   | 1.90  | 2.09  | 2.04  | 1.79  | 1.55  | 1.41  | 1.28  | 1.24  | 1.31  | 1.38  | 1.55  | 1.77 |
|                                                                               | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD18.3     | 2      | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD10.0     | 5952   | 364   | 348   | 437   | 494   | 567   | 581   | 611   | 606   | 563   | 533   | 458   | 389  |
|                                                                               | CDD18.3     | 2913   | 107   | 116   | 179   | 244   | 309   | 331   | 352   | 348   | 313   | 274   | 208   | 131  |
|                                                                               | CDH23.3     | 28026  | 159   | 299   | 870   | 2063  | 3486  | 4181  | 4591  | 4430  | 3746  | 2658  | 1292  | 253  |
|                                                                               | CDH26.7     | 9978   | 8     | 20    | 123   | 525   | 1232  | 1841  | 2136  | 1981  | 1421  | 554   | 126   | 12   |
| Wind                                                                          | WSAvg       | 5.4    | 7.3   | 6.2   | 5.2   | 4.2   | 3.7   | 3.7   | 3.9   | 4.0   | 4.5   | 6.5   | 7.5   | 8.1  |
| Precipitation                                                                 | PrecAvg     | 1065   | 29    | 17    | 17    | 28    | 119   | 163   | 110   | 184   | 190   | 119   | 46    | 49   |
|                                                                               | PrecMax     | 2452   | 222   | 67    | 106   | 198   | 349   | 416   | 276   | 616   | 978   | 603   | 185   | 325  |
|                                                                               | PrecMin     | 232    | 0     | 0     | 0     | 0     | 8     | 0     | 8     | 0     | 14    | 0     | 0     | 0    |
|                                                                               | PrecStd     | 522    | 48    | 19    | 23    | 42    | 90    | 118   | 65    | 152   | 191   | 160   | 49    | 67   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.6  | 28.4  | 30.1  | 32.3  | 33.5  | 34.0  | 34.5  | 34.7  | 33.6  | 32.0  | 30.3  | 28.2 |
|                                                                               |             | MCWB   | 23.9  | 25.2  | 26.3  | 27.6  | 28.7  | 29.5  | 29.1  | 28.3  | 28.5  | 27.7  | 27.1  | 24.4 |
|                                                                               | 2%          | DB     | 26.2  | 27.2  | 29.0  | 31.4  | 32.7  | 33.1  | 33.6  | 33.3  | 32.6  | 30.8  | 29.1  | 26.5 |
|                                                                               |             | MCWB   | 23.0  | 24.3  | 25.8  | 27.3  | 28.3  | 29.2  | 29.0  | 28.6  | 28.3  | 26.9  | 26.2  | 23.5 |
|                                                                               | 5%          | DB     | 24.9  | 26.2  | 28.0  | 30.5  | 32.0  | 32.6  | 32.9  | 32.6  | 31.8  | 29.9  | 28.0  | 25.4 |
|                                                                               |             | MCWB   | 22.2  | 23.6  | 25.0  | 26.8  | 28.0  | 28.9  | 28.6  | 28.6  | 27.9  | 26.5  | 25.4  | 22.9 |
|                                                                               | 10%         | DB     | 24.0  | 25.1  | 27.1  | 29.5  | 31.2  | 32.0  | 32.3  | 32.0  | 31.1  | 29.1  | 27.2  | 24.6 |
|                                                                               |             | MCWB   | 21.7  | 22.8  | 24.4  | 26.2  | 27.7  | 28.6  | 28.4  | 28.5  | 27.5  | 26.1  | 24.9  | 22.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 24.7  | 25.8  | 27.7  | 30.0  | 29.7  | 31.2  | 31.7  | 31.3  | 30.5  | 28.4  | 28.1  | 25.1 |
|                                                                               |             | MCDB   | 26.4  | 27.7  | 28.9  | 30.8  | 31.6  | 32.1  | 32.8  | 32.2  | 31.6  | 30.7  | 29.5  | 27.0 |
|                                                                               | 2%          | WB     | 23.6  | 24.7  | 26.1  | 28.2  | 29.0  | 29.7  | 29.5  | 29.7  | 28.7  | 27.7  | 26.7  | 24.2 |
|                                                                               |             | MCDB   | 25.3  | 26.5  | 28.3  | 30.3  | 31.7  | 32.3  | 32.2  | 31.8  | 31.5  | 29.9  | 28.2  | 25.9 |
|                                                                               | 5%          | WB     | 22.7  | 24.0  | 25.3  | 27.1  | 28.5  | 29.2  | 29.0  | 28.9  | 28.2  | 27.1  | 26.1  | 23.4 |
|                                                                               |             | MCDB   | 24.4  | 25.6  | 27.5  | 29.8  | 31.1  | 31.9  | 32.1  | 31.8  | 31.0  | 29.2  | 27.4  | 25.0 |
|                                                                               | 10%         | WB     | 22.0  | 23.2  | 24.7  | 26.4  | 28.0  | 28.7  | 28.6  | 28.5  | 27.8  | 26.6  | 25.4  | 22.7 |
|                                                                               |             | MCDB   | 23.7  | 24.7  | 26.6  | 28.9  | 30.5  | 31.4  | 31.5  | 31.3  | 30.4  | 28.6  | 26.8  | 24.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 3.2   | 3.6   | 4.3   | 4.8   | 4.7   | 4.5   | 4.5   | 4.5   | 4.2   | 3.2   | 2.7   | 2.7  |
|                                                                               |             | MCDBR  | 4.9   | 5.3   | 5.8   | 6.2   | 5.7   | 5.1   | 5.5   | 5.5   | 5.3   | 4.4   | 4.0   | 4.2  |
|                                                                               |             | MCWBR  | 3.6   | 3.5   | 3.4   | 3.4   | 2.8   | 2.6   | 2.8   | 2.7   | 2.8   | 2.7   | 2.7   | 3.1  |
|                                                                               | 5% WB       | MCDBR  | 4.9   | 5.1   | 5.5   | 5.9   | 5.3   | 4.8   | 5.0   | 5.0   | 4.8   | 4.1   | 3.7   | 3.9  |
|                                                                               |             | MCWBR  | 3.8   | 3.6   | 3.6   | 3.8   | 3.1   | 2.8   | 2.8   | 2.9   | 2.7   | 2.8   | 2.6   | 3.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.539  | 0.548 | 0.643 | 0.602 | 0.483 | 0.472 | 0.471 | 0.506 | 0.533 | 0.578 | 0.570 | 0.587 |      |
|                                                                               | taud        | 1.944  | 1.943 | 1.764 | 1.900 | 2.302 | 2.351 | 2.349 | 2.223 | 2.108 | 1.927 | 1.927 | 1.844 |      |
|                                                                               | Ebn at Noon | 728    | 753   | 700   | 738   | 823   | 825   | 827   | 803   | 776   | 718   | 695   | 666   |      |
|                                                                               | Edh at Noon | 174    | 184   | 228   | 201   | 133   | 126   | 126   | 144   | 159   | 184   | 176   | 187   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.85   | 3.69  | 4.51  | 5.49  | 5.78  | 5.81  | 5.95  | 5.36  | 5.00  | 4.37  | 3.21  | 2.55  |      |
|                                                                               | RadStd      | 0.52   | 0.71  | 0.43  | 0.56  | 0.46  | 0.57  | 0.44  | 0.55  | 0.59  | 0.44  | 0.43  | 0.46  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page